

Practice Question of Discreet Structure

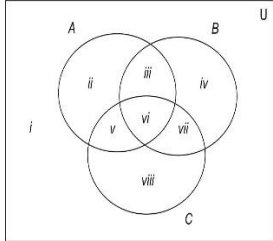
Question No 01:

Is each of the following a well-defined set? Give brief reasons for each of your answers.

- The collection of all alphanumeric characters.
- The collection of all tall people.
- The collection of all real numbers x for which:
 $2x - 9 = 16$.
- The collection of all integers x for which:
 $2x - 9 = 16$.
- The collection of all good tennis players.

Question No 02:

- Copy the truth table to the right and write at the end of each row the number of the corresponding region in Fig. 4 Venn Diagrams.



In A?	In B?	In C?	Region
Y	Y	Y	
Y	Y	N	
Y	N	Y	
Y	N	N	
N	Y	Y	
N	Y	N	
N	N	Y	
N	N	N	

Question No 03:

If $U = \{\text{letters of the alphabet}\}$, $A = \{a, a, a, b, b, a, c\}$, $B = \{c, b, a, c\}$ and $C = \{a, b, c\}$, what can be said about A, B and C?

Question No 04:

$U = \{\text{natural numbers}\}$; $A = \{2, 4, 6, 8, 10\}$; $B = \{1, 3, 6, 7, 8\}$ State whether each of the following is true or false:

- $A \subset U$
- $B \subseteq A$
- $\emptyset \subset U$

Question No 05:

$U = \{a, b, c, d, e, f, g, h\}$; $P = \{c, f\}$; $Q = \{a, c, d, e, f, h\}$; $R = \{c, d, h\}$

- Draw a Venn diagram, showing these sets with all the elements entered into the appropriate regions. If necessary, redraw the diagram to eliminate any empty regions.
- Which of sets P, Q and R are proper subsets of others? Write your answer(s) using the $\subset$ symbol.
- P and R are disjoint sets. True or False?

Question No 06:

Sketch Venn diagrams that show the universal set, U, the sets A and B, and a single element x in each of the following cases:

- $x \in A$; $A \subset B$
- $x \in A$; A and B are disjoint
- $x \in A$; $x \notin B$; $B \subset A$
- $x \in A$; $x \in B$; A is not a subset of B; B is not a subset of A

Question No 05:

$$U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

$$A = \{2, 4, 6, 8, 10\}$$

$$B = \{1, 3, 6, 7, 8\}$$

$$C = \{3, 7\}$$

Part 1:

(a) Illustrate the sets U, A, B and C in a Venn diagram, marking all the elements in the appropriate places. (Note: if any region in your diagram does not contain any elements, re-draw the set loops to correct this.)

(b) Using your Venn diagram, list the elements in each of the following sets:

$$A \cap B, A \cup C, A', B', B \cap A', B \cap C', A - B, A \Delta B$$

(c) Complete the statement using a single symbol: $C - B =$

Part 2:

True or false?

(a) $|\emptyset| = 1$

(b) $|\{x, x\}| = 2$

(c) $|U \cap \emptyset| = 0$

Part 3:

What can you say about two sets P and Q if:

(a) $P \cap Q' = \emptyset$

(b) $P \cup Q = P$?

Part 4:

Make six copies of the Venn diagram shown alongside, and then shade the areas represented by:

(a) $A' \cup B$

(b) $A \cap B'$

(c) $(A \cap B)'$

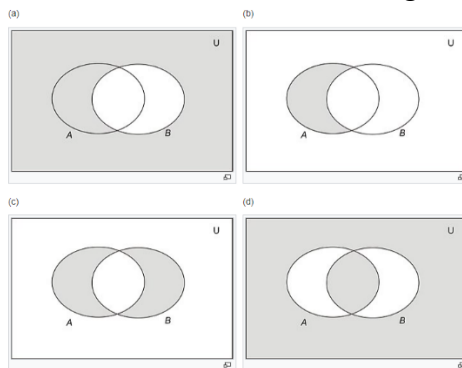
(d) $A' \cup B'$

(e) $(A \cup B)'$

(f) $A' \cap B'$

Part 5:

Identify the sets represented by each of the shaded areas below, using the set notation symbols $\cap$, $\cup$ and $'$ only:

**Part 6:**

(a) One of the shaded regions in question 5 represents the set $A - B$. Identify which one it is, and hence write a definition of $A - B$ using only symbols from the list $\cap$, $\cup$ and $'$.

(b) Again using one of your answers to question 5, write a definition of $A \Delta B$ using only symbols from the list $\cap$, $\cup$ and $'$. (There are two possibilities here – see if you can find them both!)

Power Set:

Question No 01:

- (a) If $A = \{1, 2, 3, 4\}$, write down $P(A)$ by listing its elements. What is the value of $|P(A)|$?
(b) If $|A| = 5$, what is the value of $|P(A)|$?
(c) If $|A| = 10$, what is the value of $|P(A)|$?

Question No 02:

Prove the following identities, stating carefully which of the set laws you are using at each stage of the proof.

- (a) $B \cup (\emptyset \cap A) = B$
(b) $(A' \cap U)' = A$
(c) $(C \cup A) \cap (B \cup A) = A \cup (B \cap C)$
(d) $(A \cap B) \cup (A \cap B') = A$
(e) $(A \cap B) \cup (A \cup B')' = B$
(f) $A \cap (A \cup B) = A$

Cartesian product

Question No 01:

$$X = \{a, c\} \text{ and } Y = \{a, b, e, f\}.$$

Write down the elements of:

- (a) $X \times Y$
(b) $Y \times X$
(c) $X^2 (= X \times X)$
(d) What could you say about two sets A and B if $A \times B = B \times A$?

Question No 02:

A chess board's 8 rows are labelled 1 to 8, and its 8 columns a to h. Each square of the board is described by the ordered pair (column letter, row number).

- (a) A knight is positioned at (d, 3). Write down its possible positions after a single move of the knight.
(b) If $R = \{1, 2, \dots, 8\}$, $C = \{a, b, \dots, h\}$, and $P = \{\text{coordinates of all squares on the chess board}\}$, use set notation to express P in terms of R and C .
(c) A rook is positioned at (g, 2). If $T = \{2\}$ and $G = \{g\}$, express its possible positions after one move of the rook in terms of R , C , T and G .

Question No 03:

In a certain programming language, all variable names have to be 3 characters long. The first character must be a letter from 'a' to 'z'; the others can be letters or digits from 0 to 9.

If $L = \{a, b, c, \dots, z\}$, $D = \{0, 1, 2, \dots, 9\}$, and $V = \{\text{permissible variable names}\}$, use a Cartesian product to complete:

$$V = \{pqr \mid (p, q, r) \in \dots\}$$

Question No 04:

It is believed that, for any sets A , B and C , $A \times (B \cap C) = (A \times B) \cap (A \times C)$. (Note that, if this is true, it says that $\times$ is distributive over $\cap$.) Copy and complete the two Cartesian diagrams shown below – one for the expression on each side of the equation – to investigate this. Do you think that the proposition is true?